

Implementation of CUMAV learning with rate curves in ACEIT

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1 BLUF

Within the ACEIT suite, there are three implementations of cumulative average (CUMAV) learning with rate curves (LRCs):

1. CO\$TAT direct method (CUMAV-D),
2. CO\$TAT iterative method (CUMAV-I),
3. ACE R phasing method.

While (2) and (3) are equivalent, (1) is not equivalent to (2) or (3). This is problematic, since (1) is considered the default method for fitting CUMAV LRCs in CO\$TAT. Numerical experiments show that the discrepancy between (1) and (2) and (3) can be practically significant under realistic conditions. For the sake of consistency between CO\$TAT and ACE—in addition to other statistical reasons—the use of (1) is not recommended.

2 Introduction

The *Cumulative Average (CUMAV) Learning Curve (LC)* model relates cumulative average production costs (CAC) to cumulative total production quantities (CTQ) via a discrete power law

$$CAC = a \cdot CTQ^b.$$

The model parameters a and b represent the cost of the first unit (T1) and control the rate at which CAC changes as more units are produced, respectively. Typically, $b < 0$ so that costs decrease as more units are produced. This is an example of a manufacturing cost improvement effect.

Another cost improvement effect is discounting based on buy quantity. This is modeled by introducing an additional power law term associated with the production rate ∂CTQ^r . Here, ∂CTQ represents the production rate and r is a parameter controlling the rate at which CAC changes with production rate. Similarly to b , we usually take $r < 0$ so that costs decrease with increasing production rate¹. In the setting where production is broken up into sequential lots, lot quantities are often used as a proxy for the production rate. The learning curve with additional production rate term is called a *learning with rate curve (LRC)*.

Both CO\$TAT and ACE implement CUMAV LRCs. CO\$TAT provides two distinct methods, *direct (CUMAV-D)* and *iterative (CUMAV-I)* for fitting CUMAV LRCs. Of the two, CUMAV-D is usually considered the default method. The key finding of this document is that the incorporation of the production rate term into the base CUMAV LC is equivalent for CUMAV-I and ACE, while different for CUMAV-D. This results in discrepancies in fitted CACs obtained from CUMAV-D and CACs calculated within ACE. In the subsequent sections, we will identify the underlying CUMAV LRC models used by CO\$TAT and ACE, write down an expression for the discrepancy between the two, and conduct numerical experiments showcasing the discrepancy under realistic conditions.

¹LRCs are often parametrized by $B = 2^b$ and $R = 2^r$. B and R are called the learning and rate *slopes*, respectively, and are often expressed as percentages. The learning and rate slopes determine the percentage change in CAC resulting from doubling CTQ and ∂CTQ , respectively. A fundamental property of power laws is that these percentages do not depend on the initial values of CTQ and ∂CTQ before doubling.

3 CUMAV LRC implementations

We will start by identifying the underlying CUMAV LRC models used by CO\$TAT and ACE. This will be based on application documentation as well as numerical verification. CO\$TAT provides two variants of the CUMAV LRC, *direct* (CUMAV-D) and *iterative* (CUMAV-I). We believe that ACE uses CUMAV-I. Before proceeding, we will need to discuss lot based learning, as both CO\$TAT and ACE use lot based learning instead of the more general model described in Section 2.

Suppose production is broken into n sequential lots, where lot i consists of units $f_i, f_i + 1, \dots, \ell_i$. In other words, f_i is the first unit of lot i , ℓ_i is the last unit of lot i , and lot i is contiguous, including all units in between. Moreover, assume that $f_1 = 1$ and $\forall i > 1 \quad f_i = \ell_{i-1}$, so that collectively the lots account for all units produced between f_1 and ℓ_n . Noting that the CTQ through lot i is ℓ_i , the base CUMAV LC becomes

$$y_i = a\ell_i^b,$$

where y_i is the CAC through lot i . The lot quantities given by $q_i = \ell_i - f_i + 1$ serve as a proxy for the production rate ∂CTQ . Thus, the rate term takes on the form

$$q_i^r = (\ell_i - f_i + 1)^r.$$

3.1 CUMAV-D

CUMAV-D incorporates the rate effect in the obvious way. The CAC for lots 1 through i is given by

$$y_i = a\ell_i^b q_i^r.$$

This expression appears in the documentation² and is verified numerically. It is also useful to view things from the perspective of lot total cost (LTC) instead of CAC. Denoting the total cost of lot i by $y_i^\dagger$, we have

$$\begin{aligned} y_i^\dagger &= \begin{cases} \ell_i y_i - \ell_{i-1} y_{i-1} & \text{if } i > 1 \\ \ell_1 y_1 & \text{otherwise} \end{cases} \\ &= \begin{cases} a(\ell_i^{1+b} q_i^r - \ell_{i-1}^{1+b} q_{i-1}^r) & \text{if } i > 1 \\ a\ell_1^{1+b} q_1^r & \text{otherwise.} \end{cases} \end{aligned} \quad (3.1)$$

Note that $y_i^\dagger$ depends on f_i, ℓ_i as well as f_{i-1}, ℓ_{i-1} .

3.2 CUMAV-I/ACE

Next, we turn our attention to CUMAV-I, which we claim to be equivalent to the model used by ACE to populate R phased rows. CO\$TAT documentation describes CUMAV-I in terms of a lot plot point, which plays the same role in CUMAV LRCs as the lot midpoint in Unit Theory LRCs. We will simplify the discussion by not invoking the lot plot point, as it has no utility for us. The iterative model is more naturally understood in terms of LTC. CUMAV-I models the total cost of lot i as

$$z_i^\dagger = \begin{cases} a(\ell_i^{1+b} - \ell_{i-1}^{1+b}) q_i^r & \text{if } i > 1 \\ a\ell_1^{1+b} q_1^r & \text{otherwise.} \end{cases} \quad (3.2)$$

Comparing $z_i^\dagger$ with $y_i^\dagger$, there is a key difference in that $z_i^\dagger$ depends only on the data of lot i , while $y_i^\dagger$ depends on that data of lot i and $i - 1$. We can calculate CAC from LTC by

$$z_i = \frac{1}{\ell_i} \sum_{j=1}^i z_j^\dagger,$$

though the explicit form is complicated and depends on the data of lots 1 through i . On the other hand, y_i depends only on the data of lot i .

²See the CO\$TAT Help page titled *Learning Theory*.

As for the equivalence of CUMAV-I and ACE, the ACE documentation is less helpful. We could not locate documentation explicitly describing how ACE incorporates the rate effect into the CUMAV LC. However, we have observed the coincidence of CUMAV-I and ACE numerically. There are also some other small clues. For instance, ACE allows first and last units to be provided only for lots $i > m$ for some $m > 0$. The only data on lots $1, \dots, m$ required by ACE is the prior quantity ℓ_m . This would make it impossible to calculate $y_{m+1}^\dagger$, which depends on f_m . On the other hand, $z_{m+1}^\dagger$ depends only on f_{m+1}, ℓ_{m+1} and could be calculated.

3.3 Discrepancy between CUMAV-D and CUMAV-I/ACE

We will compare LTCs obtained from the CUMAV-D model against the CUMAV-I/ACE model. To simplify the analysis, we will further assume that $-1 < b < 0$ and $r < 0$. The constraints $b, r < 0$ force the models to decrease costs as CTQ and production rate increase. The $-1 < b$ is needed to avoid nonpositive LTCs. In general, values of b even approaching -1 would be highly unrealistic, so this constraint is fairly innocent. Clearly, $y_1^\dagger - z_1^\dagger = 0$, and for $i > 1$, we have

$$\begin{aligned} y_i^\dagger - z_i^\dagger &= a (\ell_i^{1+b} q_i^r - \ell_{i-1}^{1+b} q_{i-1}^r) - a (\ell_i^{1+b} - \ell_{i-1}^{1+b}) q_i^r \\ &= a \ell_{i-1}^{1+b} (q_i^r - q_{i-1}^r). \end{aligned} \quad (3.3)$$

We find that the magnitude of the discrepancy increases with $\ell_{i-1} = f_i - 1$ as well as the change in lot quantity going from q_{i-1} to q_i . A numerical experiment will demonstrate that this discrepancy can be practically significant.

4 Numerical experiments

We will conduct a numerical experiment using CO\$STAT and ACE version 8.2. The experiment will be based on the hypothetical quantity stream in Table 1. For each lot i , we calculated $y_i^\dagger$ and $z_i^\dagger$ using (3.1) and (3.2) with $a = 1, b = -0.1, r = -0.05$ ³. We then fit CUMAV-D and CUMAV-I to the data in CO\$STAT. As expected, CUMAV-D fit $y^\dagger$ exactly, and CUMAV-I fit $z^\dagger$ exactly. This provides numerical verification that (3.1) and (3.2) are correct. We then created an R phased row in ACE and input our quantity stream and LRC parameters. ACE results matched $z^\dagger$ exactly⁴. The same calculations were repeated with $a = 1, b = -0.1, r = -0.1$. The outcome of the second experiment was the same.

In both experiments, the discrepancy between $y_i^\dagger$ and $z_i^\dagger$ tracked variability in the quantity stream and generally increased as CTQ increased, as predicted by (3.3). In particular, the discrepancy was 0 for $i = 1$ and $i = 12$. The total percent discrepancy $100\% \cdot (\sum_i y_i^\dagger - \sum_i z_i^\dagger) / \sum_i z_i^\dagger$ was 7.4% for the first experiment and 15.2% for the second.

5 Conclusion

We have identified two distinct CUMAV LRC models within the ACEIT suite, CUMAV-D and CUMAV-I. Both are implemented in CO\$STAT, while ACE uses CUMAV-I to populate R phased rows. Fitting a CUMAV-D model in CO\$STAT and transferring the fitted parameters to ACE results in a discrepancy between fitted values in CO\$STAT and ACE results. We have shown that the discrepancy in LTC grows with increasing CTQ and lot to lot variability in lot quantities. Numerical experiments are consistent with the postulated expressions for CUMAV-D and CUMAV-I, as well as the use of CUMAV-I within ACE. Moreover, they show that the discrepancy can be practically significant.

In practice, CUMAV-D is often considered the default CUMAV model in CO\$STAT. This is unfortunate both due to the aforementioned discrepancy between CO\$STAT and ACE, but also certain statistical issues surrounding CUMAV-D. Table 1 and Table 2 show that CUMAV-D can exhibit unrealistic patterns in LTC relative to quantities. Nonetheless, in addition to being the first option in the CO\$STAT menu, CUMAV-D often exhibits superior statistical metrics such as R^2 , standard errors, coefficient of variation, t-statistic, etc. This can lead analysts to choose CUMAV-D when other alternatives (e.g. CUMAV-I or unit theory)

³We do not add any random noise, as we are concerned only with the deterministic aspects of LRCs.

⁴The maximum precision within ACE is 6 decimal places. By “exactly”, we mean up to the 6th decimal place.

Table 1: Data set for numerical experiment with $a = 1, b = -0.1, r = -0.05$.

i	q_i	f_i	ℓ_i	$y_i^\dagger$	$z_i^\dagger$	$100\% \cdot (y_i^\dagger - z_i^\dagger)/z_i^\dagger$
1	5	1	5	3.927575	3.927575	0.0%
2	60	6	65	30.963509	31.422394	-1.5%
3	100	66	165	43.765585	44.645464	-2.0%
4	112	166	277	46.015895	46.460337	-1.0%
5	107	278	384	42.987224	42.702209	0.7%
6	98	385	482	38.964338	38.226177	1.9%
7	103	483	585	38.732854	39.246311	-1.3%
8	151	586	736	50.608585	55.257063	-8.4%
9	145	737	881	52.700690	52.100069	1.2%
10	148	882	1029	51.887448	52.244274	-0.7%
11	150	1030	1179	51.883034	52.151776	-0.5%
12	150	1180	1329	51.490601	51.490602	0.0%
13	100	1330	1429	45.017879	34.697370	29.7%
14	59	1430	1488	35.575673	20.900878	70.2%
15	29	1489	1515	33.215493	9.917134	234.9%

Table 2: Data set for numerical experiment with $a = 1, b = -0.1, r = -0.1$.

i	q_i	f_i	ℓ_i	$y_i^\dagger$	$z_i^\dagger$	$100\% \cdot (y_i^\dagger - z_i^\dagger)/z_i^\dagger$
1	5	1	5	3.623898	3.623898	0.0%
2	60	6	65	24.808068	25.605408	-3.1%
3	100	66	165	34.047247	35.463152	-4.0%
4	112	166	277	35.992159	36.696231	-1.9%
5	107	278	384	34.255764	33.805016	1.3%
6	98	385	482	31.566116	30.394819	3.9%
7	103	483	585	30.312896	31.128413	-2.6%
8	151	586	736	35.693003	42.997078	-17.0%
9	145	737	881	41.558473	40.622804	2.3%
10	148	882	1029	40.137398	40.693553	-1.4%
11	150	1030	1179	40.175741	40.594251	-1.0%
12	150	1180	1329	40.079602	40.079602	0.0%
13	100	1330	1429	43.792320	27.561101	58.9%
14	59	1430	1488	40.670799	17.045978	138.6%
15	29	1489	1515	47.170344	8.410426	460.9%

fail to achieve satisfactory metrics. Even when alternatives achieve acceptable metrics, direct comparison of metrics across alternatives often results in a decisive victory for CUMAV-D. The apparent superiority of CUMAV-D in many practical settings is, however, an illusion.

LRCs are generally fit on logarithmic scale ⁵. The response variable of CUMAV-D is the logarithm of CAC, while the response variable of CUMAV-I and unit theory (UT) is the logarithm of LTC ⁶. CO\$TAT generates metrics on logarithmic scale using the respective responses of each model. The fact that the response of CUMAV-D differs from that of CUMAV-I and UT means that metrics are not directly comparable between them. It is also worth noting that metrics for CUMAV-D are generated using standard linear regression formulas that assume that observations of the response are homoscedastic and uncorrelated. However, since the response is the logarithm of a sequence of cumulative averages, significant heteroscedasticity and serial correlation is expected. This leads to biased (generally downward) standard error estimation, which flows through to coefficient of variation and null hypothesis test statistics. Thus, while CUMAV-D tends to paint a rosy picture in CO\$TAT with various statistical metrics, those metrics are not directly comparable with alternative models or even reliable in isolation.

In light of the incompatibility of CUMAV-D with ACE and its other statistical issues, the author recommends against using it. Rather, CUMAV-I or UT should be used by default, depending on one's taste. If one's default fails to deliver a satisfactory fit, the other model may be considered, but is unlikely to yield much improvement. Indeed, CUMAV-I can be viewed (up to a reparametrization) as a 1-term Euler Maclaurin approximation of UT ⁷. This approximation is somewhat coarse for low CTQ, but improves as CTQ increases. The takeaway is that one should not fret over choosing between CUMAV-I or UT, but CUMAV-D should be avoided.

⁵CO\$TAT does provide a Minimum Unbiased Percent Error (MUPE) option which fits LRCs on unit scale. We do not consider MUPE in this document, but the same basic phenomenon are expected to occur.

⁶Some take the response to be the logarithm of lot average cost (LAC), but this has no effect on parameter estimation or the resulting statistical metrics.

⁷The *six term formula* of Hu & Smith is obtained by adding two additional terms from the Euler Maclaurin formula.